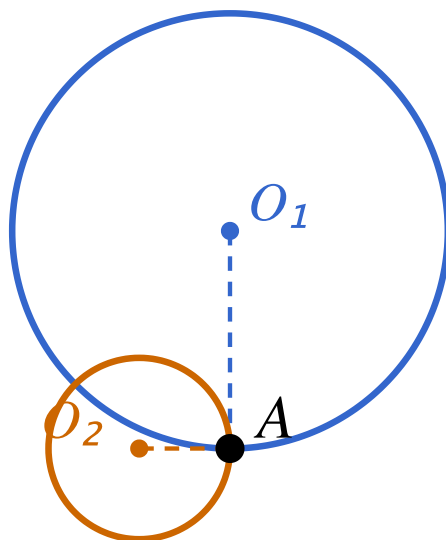


Lecture 15: Topics in Circles

Part 1: How Many Circles Pass Through Given Points?

One Point

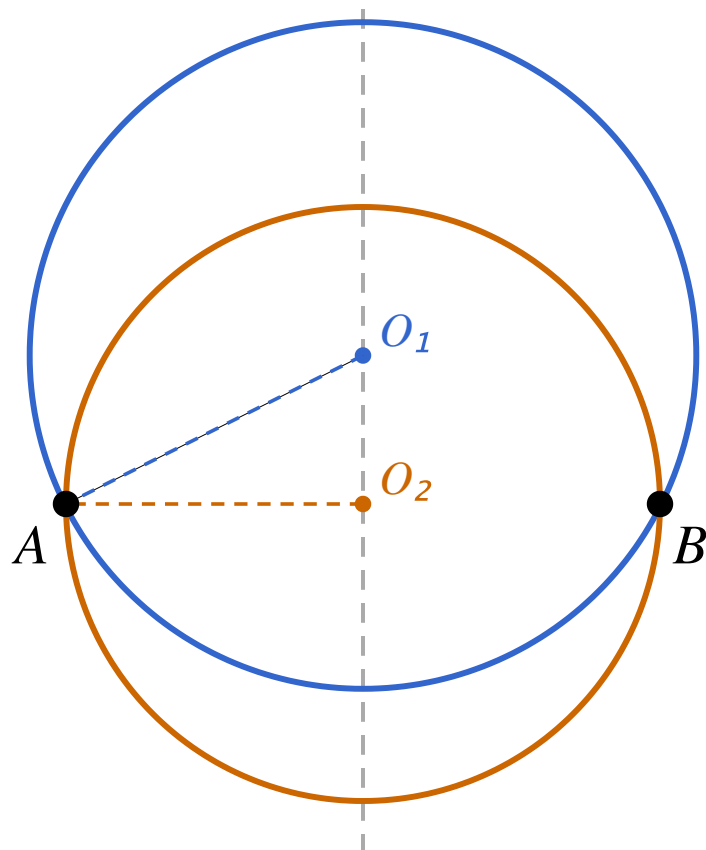
Given a single point A , infinitely many circles pass through A : we can choose any center and any radius as long as the circle passes through A .



Any center O and radius OA gives a circle through A .

Two Points

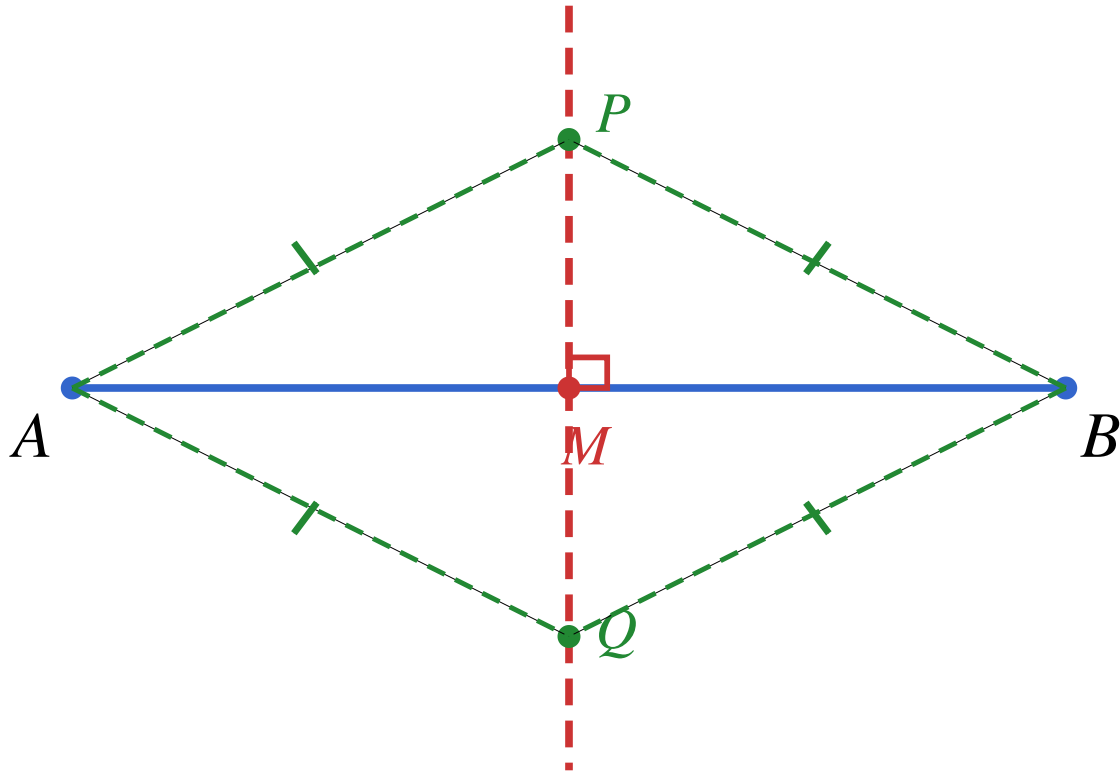
Given two distinct points A and B , we can still draw infinitely many circles through both. Every such circle's center must be **equidistant** from A and B , so the centers all lie on a special line.



All centers lie on the perpendicular bisector of AB .

This motivates the following definition.

Definition (Perpendicular Bisector): The **perpendicular bisector** of a segment AB is the line that passes through the midpoint M of AB and is perpendicular to AB .



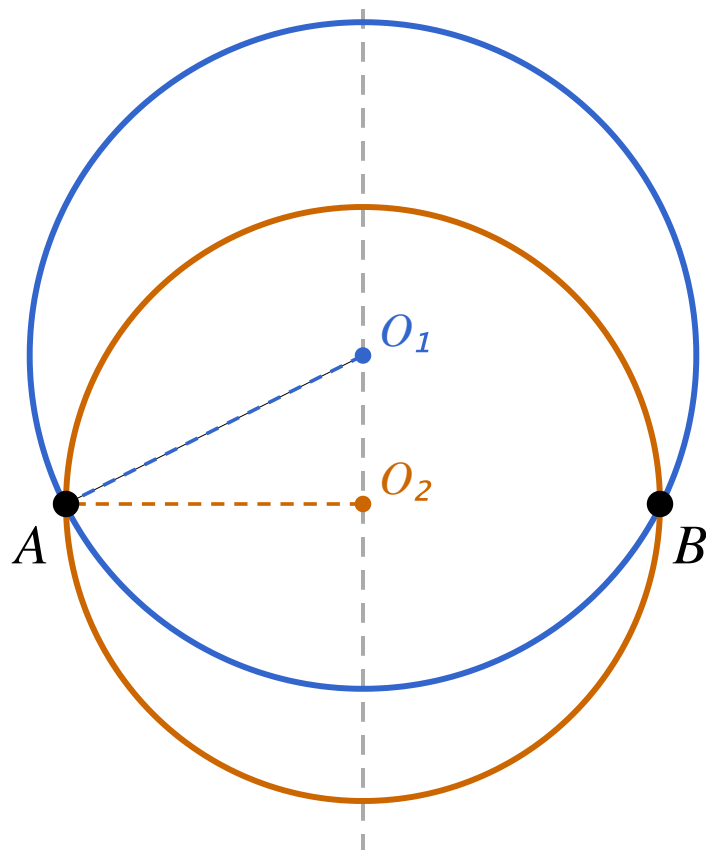
Perpendicular bisector of AB : $PA = PB$ for all P on the line

Lemma: A point P lies on the perpendicular bisector of AB if and only if $PA = PB$.

Proof:

- ($\Rightarrow$) Suppose P is on the perpendicular bisector. Then $\triangle PMA \cong \triangle PMB$ by **SAS** (PM shared, $\angle PMA = \angle PMB = 90^\circ$, $MA = MB$). Hence $PA = PB$.
- ($\Leftarrow$) Suppose $PA = PB$. Then $\triangle PMA$ and $\triangle PMB$ share PM , have $MA = MB$, and $PA = PB$. By **SSS**, the two triangles are congruent, so $\angle PMA = \angle PMB$. Since they are supplementary, $\angle PMA = 90^\circ$, meaning P lies on the perpendicular bisector. ■

Corollary: Given two points A and B , the centers of all circles passing through both A and B lie exactly on the perpendicular bisector of AB . Hence there are infinitely many such circles.

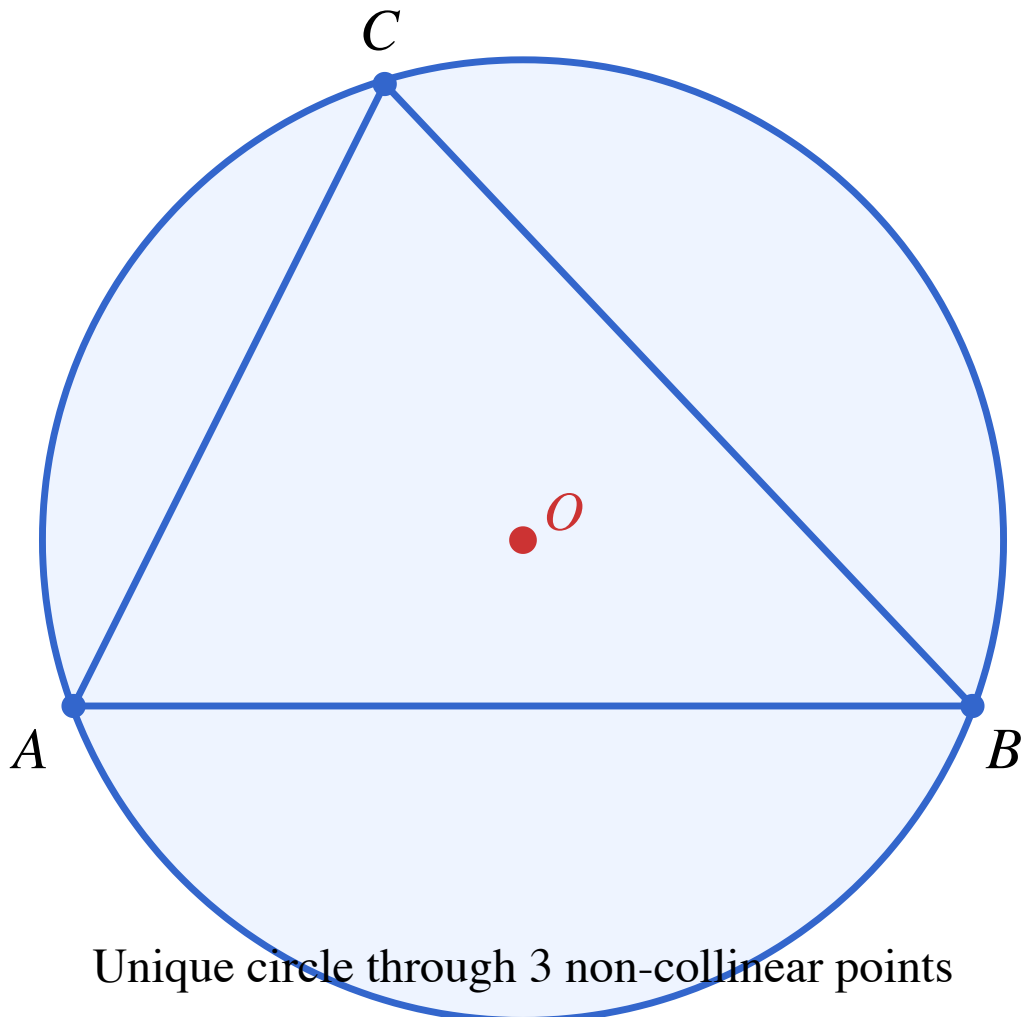


All centers lie on the perpendicular bisector of AB .

Three Points

If we are given three points A, B, C , how many circles pass through all three? If A, B, C are collinear, no circle passes through all three (a circle and a line meet in at most 2 points).

Given three **non-collinear** points A, B, C , how many circles pass through all three?

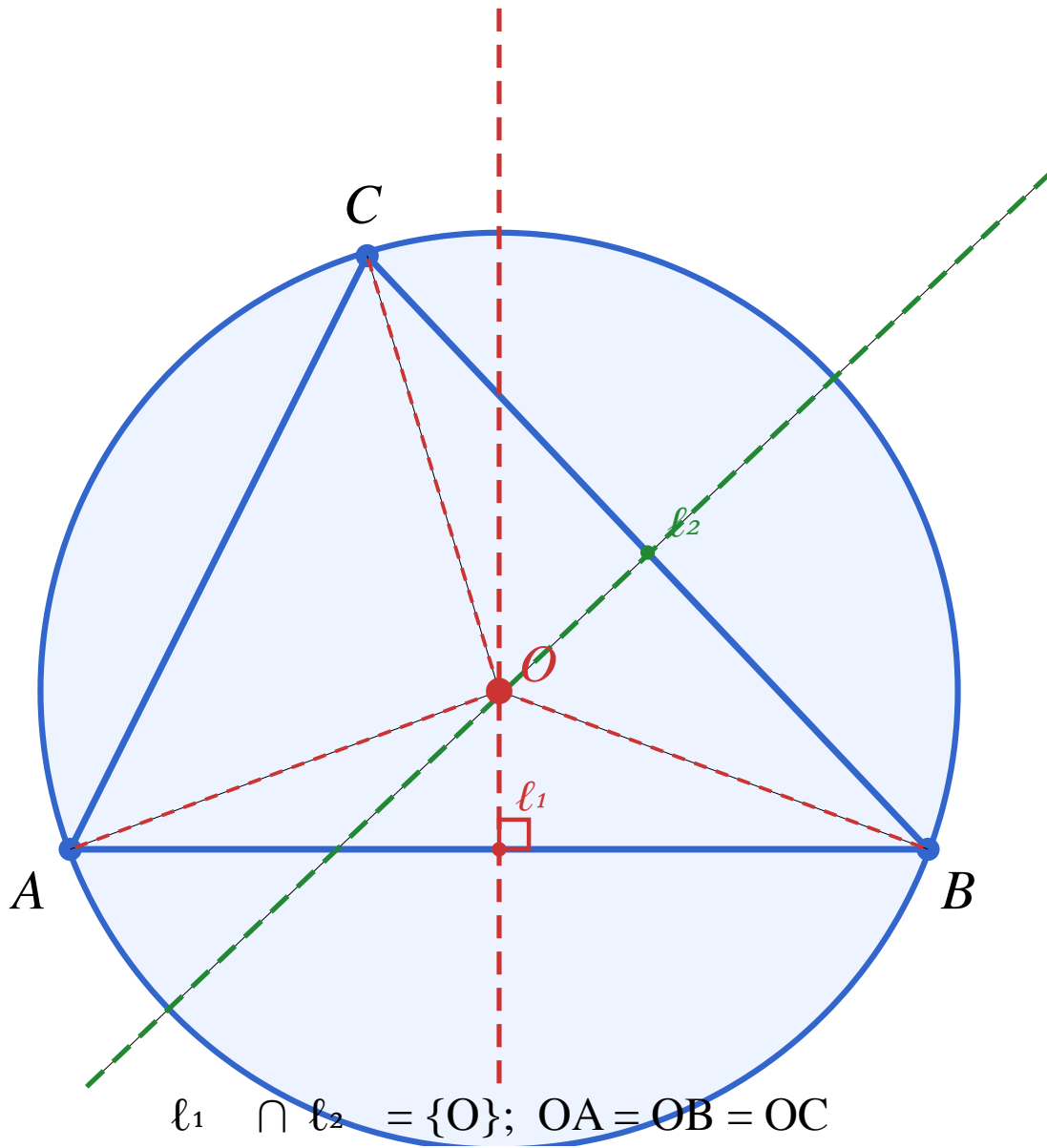


Unique circle through 3 non-collinear points

Theorem 1: Given three non-collinear points, there is exactly one circle passing through all three.

Proof:

The idea is to reduce to the two-point case. First we construct the perpendicular bisectors ℓ_1 and ℓ_2 of AB and BC . By the Lemma, any center of a circle passing through A, B, C must lie on both ℓ_1 and ℓ_2 . They intersect at a unique point O . Setting $r = OA = OB = OC$, the circle centered at O with radius r passes through A, B, C .



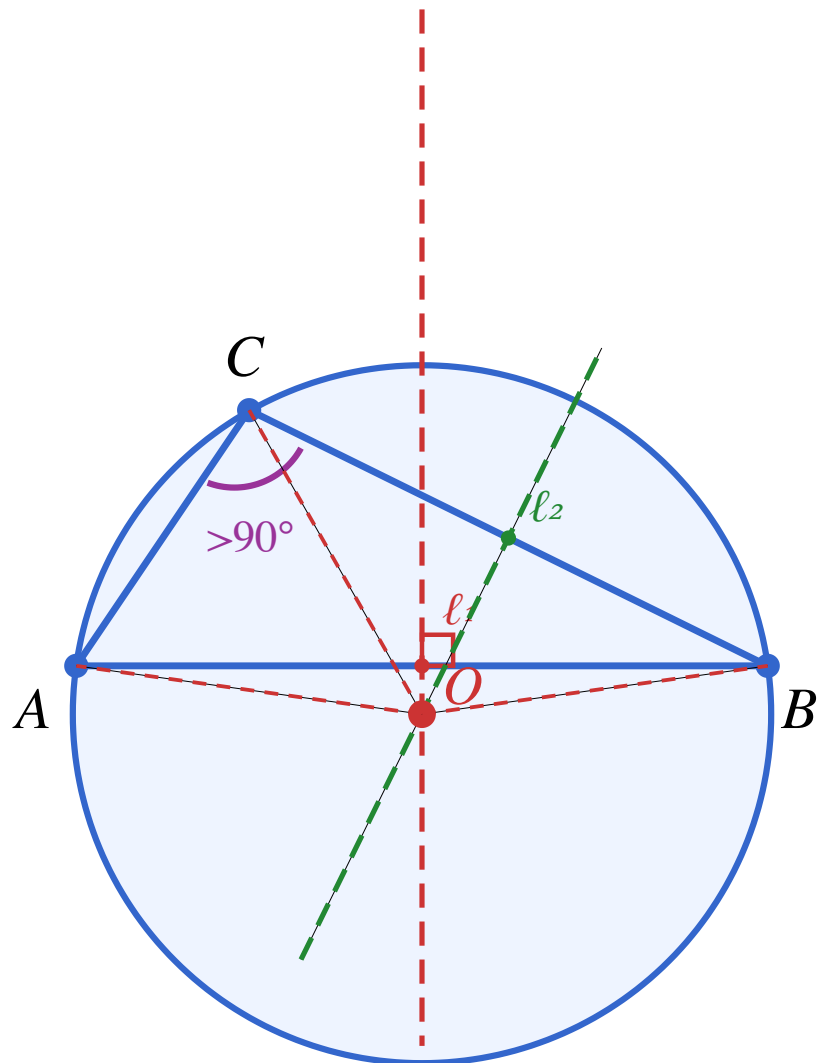
Since any center must lie on $\ell_1 \cap \ell_2 = \{O\}$, so the center and radius are determined. ■

This motivates the following definition.

Definition (Circumcenter): Given a triangle ABC , the **circumcenter** O is the intersection of the perpendicular bisectors of its three sides. The circle centered at O passing through A, B, C is the **circumscribed circle** (or **circumcircle**).

Corollary: The three perpendicular bisectors of a triangle are concurrent (they all meet at the circumcenter).

Remark: The circumcenter can lie inside the triangle (acute triangle), on the hypotenuse (right triangle), or outside the triangle (obtuse triangle).

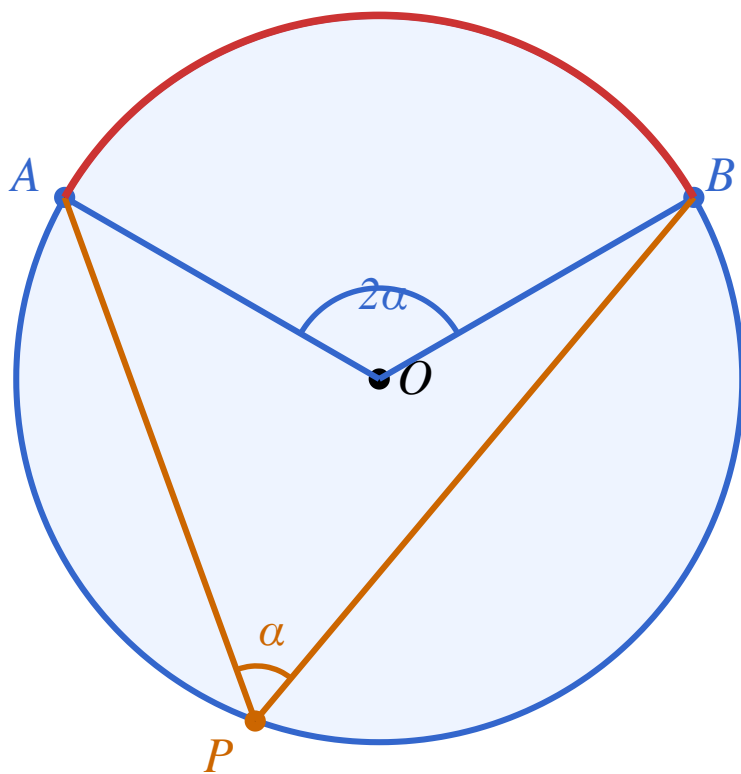


Obtuse triangle: circumcenter O lies outside the triangle

Part 2: Inscribed Angle and Central Angle

Definition: Let O be the center of a circle and A, B points on the circle.

- The **central angle** $\angle AOB$ is the angle at O subtended by the arc AB .
- An **inscribed angle** (also called a **circular angle**) $\angle APB$ is the angle at a point P on the circle outside the same arc AB .



Central angle
 $\angle AOB = 2\alpha$

Inscribed angle
 $\angle APB = \alpha$

Both subtend
 arc AB

Theorem 2 (Inscribed Angle Theorem): The inscribed angle is half the central angle of the same arc:

$$\angle APB = \frac{1}{2} \angle AOB.$$

Proof: The proof splits into three cases depending on whether the center O lies inside, on a side of, or outside the angle $\angle APB$.

Case 1 (O inside $\angle APB$): Draw the diameter PD through O . Since $OA = OP = r$, triangle OAP is isosceles, so

$$\angle OAP = \angle OPA = \beta_1.$$

The exterior angle of $\triangle OAP$ at O gives $\angle AOD = 2\beta_1$.

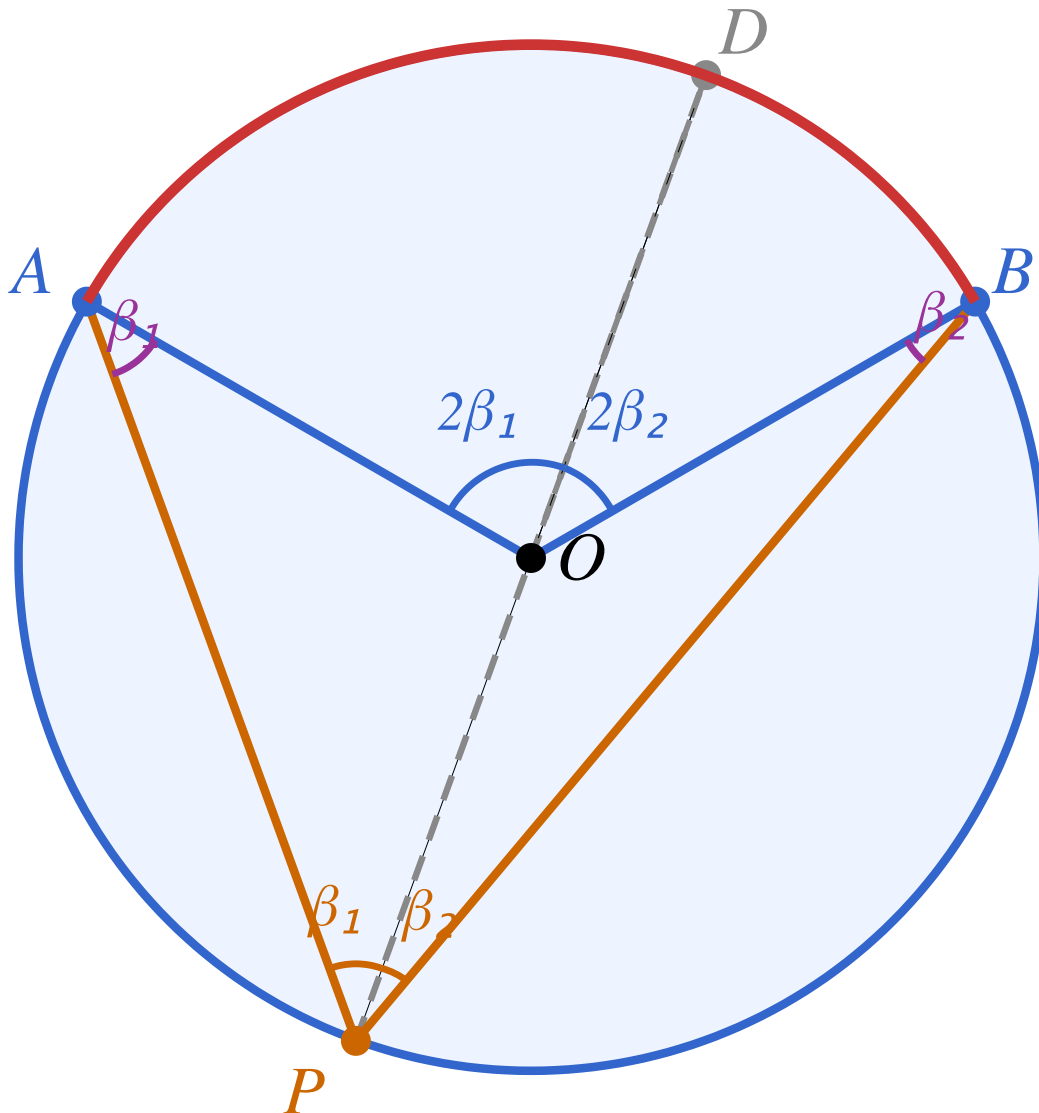
Similarly, since $OB = OP = r$, triangle OBP is isosceles, so

$$\angle OBP = \angle OPB = \beta_2,$$

and the exterior angle at O gives $\angle BOD = 2\beta_2$.

Since O is inside $\angle APB$:

$$\angle APB = \beta_1 + \beta_2 = \frac{1}{2}(2\beta_1 + 2\beta_2) = \frac{1}{2}(\angle AOD + \angle BOD) = \frac{1}{2}\angle AOB. \quad \blacksquare \text{ (Case 1)}$$



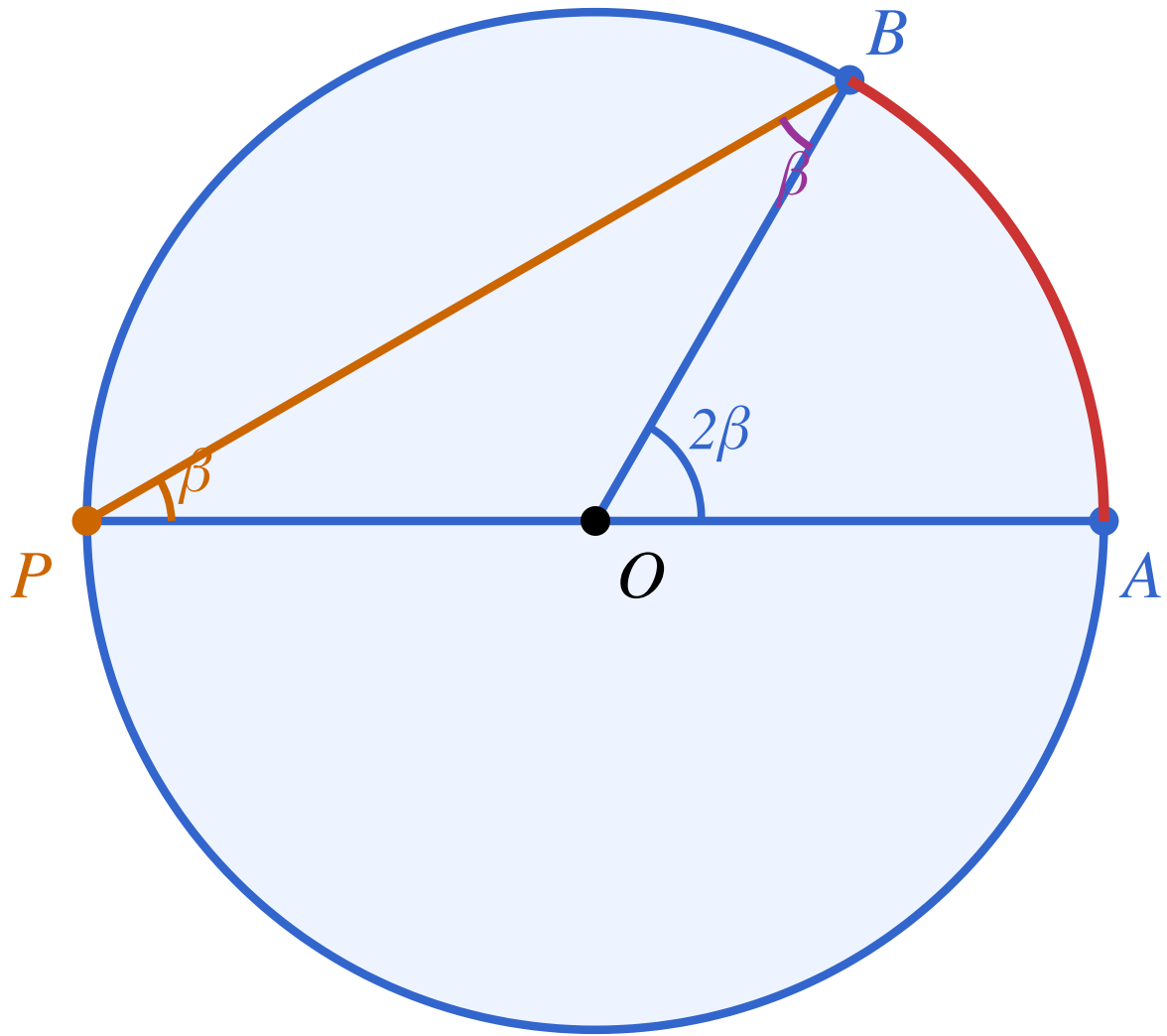
Case 1: O inside $\angle APB$; $\angle APB = \beta_1 + \beta_2 = \frac{1}{2}(2\beta_1 + 2\beta_2) = \frac{1}{2}\angle AOB$

Case 2 (O lies on PA): Here PA is a diameter, so $OA = OP = r$ (both radii). Triangle OPB is isosceles, giving

$$\angle OBP = \angle OPB = \beta.$$

The exterior angle at O satisfies $\angle BOA = 2\beta$ (exterior angle of triangle OPB). Hence

$$\angle APB = \beta = \frac{1}{2}\angle AOB. \quad \blacksquare \text{ (Case 2)}$$



Case 2: PA diameter; $\angle APB = \beta = \frac{1}{2} \angle AOB$

Case 3 (O outside $\angle APB$): Draw the diameter PD through O . Since $OA = OP = r$, triangle OAP is isosceles, so

$$\angle OAP = \angle OPA = \beta_1,$$

and the exterior angle at O gives $\angle AOD = 2\beta_1$.

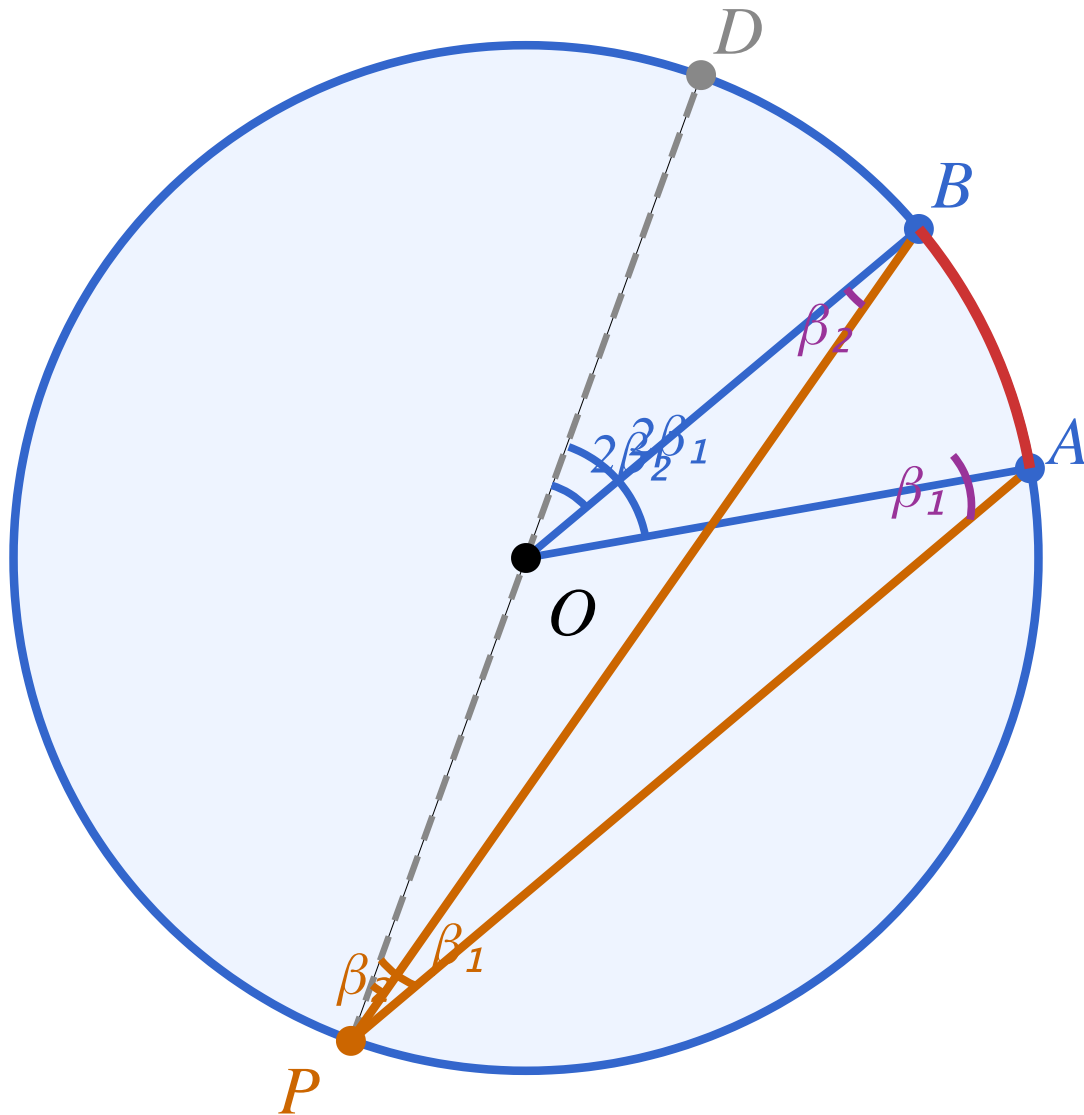
Since $OB = OP = r$, triangle OBP is isosceles, so

$$\angle OBP = \angle OPB = \beta_2,$$

and the exterior angle at O gives $\angle BOD = 2\beta_2$.

Since O is outside $\angle APB$, ray PD falls outside the angle, so $\angle APB = \angle APD - \angle BPD$:

$$\angle APB = \beta_1 - \beta_2 = \frac{1}{2}(2\beta_1 - 2\beta_2) = \frac{1}{2}(\angle AOD - \angle BOD) = \frac{1}{2}\angle AOB. \quad \blacksquare \text{ (Case 3)}$$



ase 3: O outside $\angle APB$; $\angle APB = \beta_1 - \beta_2 = \frac{1}{2}(2\beta_1 - 2\beta_2) = \frac{1}{2}\angle AOB$

Corollary 1 (Thales' Theorem): If AB is a diameter of a circle, then the inscribed angle $\angle APB = 90^\circ$ for any point P on the circle (on the semicircle not containing A or B).

Proof: The central angle $\angle AOB = 180^\circ$ (straight angle), so $\angle APB = \frac{1}{2} \cdot 180^\circ = 90^\circ$. $\blacksquare$

Corollary 2: All inscribed angles subtending the same arc are equal.

Proof: They are all equal to half the central angle subtending that arc. $\blacksquare$

Corollary 3 (Cyclic Quadrilateral): If $APBQ$ is a quadrilateral inscribed in a circle, then opposite angles are supplementary:

$$\angle APB + \angle AQB = 180^\circ.$$

Proof: P and Q lie on opposite arcs. The central angles subtended by these arcs sum to 360° , so the two inscribed angles sum to $\frac{1}{2} \cdot 360^\circ = 180^\circ$. ■